

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Marks:50

Total

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Pramith Maredukonda (192372174)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink that reads "Pramith Maredukonda". The signature is written in a cursive style with a horizontal line underneath the name.

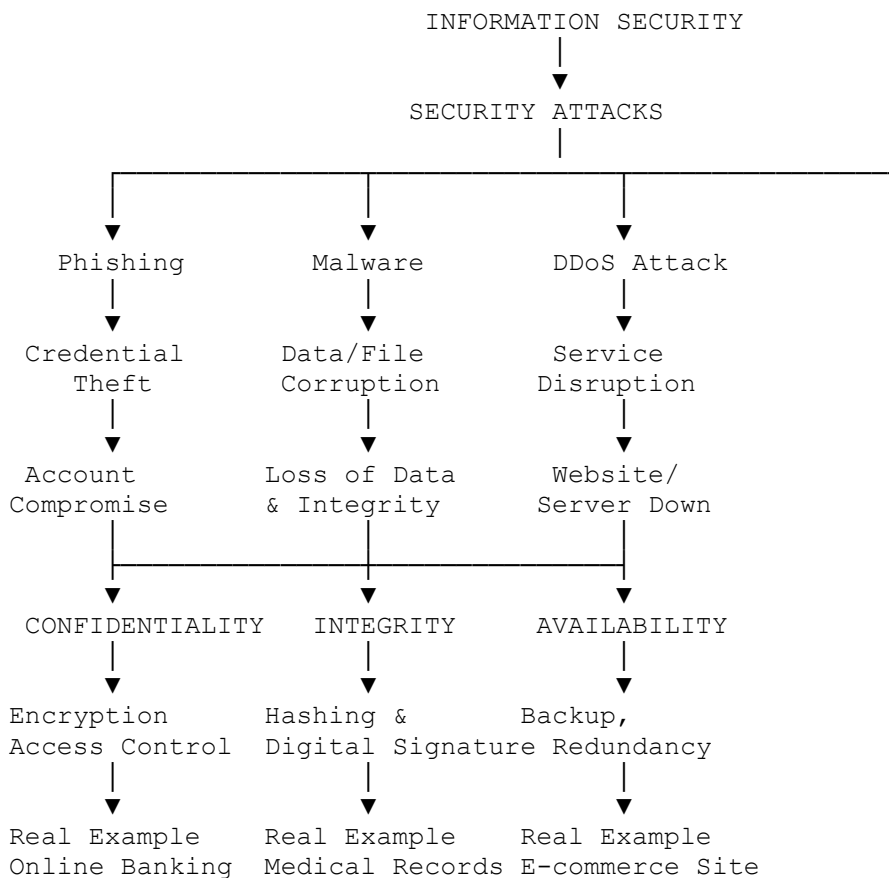
Question 1: Concept Map – Security Attacks, Threats, Risks and Security Goals

Information security is the process of protecting computer systems, networks, and data from unauthorized access, misuse, modification, or destruction. Organizations face various **security attacks** that create **threats**, which may lead to **risks** affecting valuable information and business operations. To reduce these risks, organizations implement the three primary **security goals** known as the **CIA Triad**:

- **Confidentiality**
- **Integrity**
- **Availability**
- **Authentication**

A concept map helps understand the relationship between attacks, threats, risks, and security goals.

Concept Map (Diagram)



Explanation of the Concept Map

1. Security Attacks

Security attacks are deliberate attempts to damage, steal, or gain unauthorized access to information systems.

Types of Attacks

Phishing Attack

- Attackers send fake emails or websites to steal login credentials.

Malware Attack

- Malicious software such as viruses, worms, spyware, or ransomware infects a computer.

DDoS (Distributed Denial of Service) Attack

- Multiple computers flood a server with requests, making the service unavailable.

2. Threats

A threat is a possible danger caused by a security attack.

Attack	Threat
Phishing	Credential Theft
Malware	Data Corruption
DDoS	Service Disruption

3. Risks

A risk is the possible loss or damage if a threat successfully exploits a vulnerability.

Threat	Risk
Credential Theft	Unauthorized Account Access
Data Corruption	Loss of Data Integrity
Service Disruption	Business Downtime

4. Security Goals (CIA Triad)

A. Confidentiality

Purpose:

Protects sensitive information from unauthorized access.

Security Controls

- Encryption
- Password Authentication
- Multi-Factor Authentication
- Access Control

Real-World Example

Online banking encrypts customer passwords and financial information so that only authorized users can access their accounts.

B. Integrity

Purpose:

Ensures that data remains accurate and is not modified without authorization.

Security Controls

- Hash Functions
- Digital Signatures
- Checksums
- Version Control

Real-World Example

Hospitals use digital signatures to ensure that patient medical records cannot be altered without authorization.

C. Availability

Purpose:

Ensures that systems and information remain available whenever users need them.

Security Controls

- Firewalls
- Backup Servers
- Load Balancing
- Disaster Recovery

Real-World Example

An e-commerce website uses backup servers and load balancing so customers can continue shopping even during heavy traffic or cyberattacks.

Relationship Between Attacks, Threats, Risks, and Security Goals

Security Attack	Threat	Risk	Security Goal	Real-World Example
Phishing	Credential Theft	Account Compromise	Confidentiality	Online banking login protection
Malware	File Corruption	Data Loss	Integrity	Hospital patient records
DDoS	Service Disruption	Website Downtime	Availability	Amazon or Flipkart website availability

Analysis

- Security attacks create threats that exploit vulnerabilities.
- Threats lead to risks that may affect individuals and organizations.
- The CIA Triad helps reduce these risks:
 - **Confidentiality** prevents unauthorized disclosure of data.
 - **Integrity** ensures data accuracy and prevents tampering.
 - **Availability** ensures continuous access to systems and services.
- Applying appropriate security controls significantly reduces the impact of cyberattacks.

Conclusion

Security attacks such as phishing, malware, and DDoS create different threats and risks to information systems. Organizations mitigate these risks by implementing the **CIA Triad—Confidentiality, Integrity, and Availability**—using techniques such as encryption, digital signatures, access control, backups, and disaster recovery. This relationship demonstrates how effective security goals protect information and maintain secure communication systems.

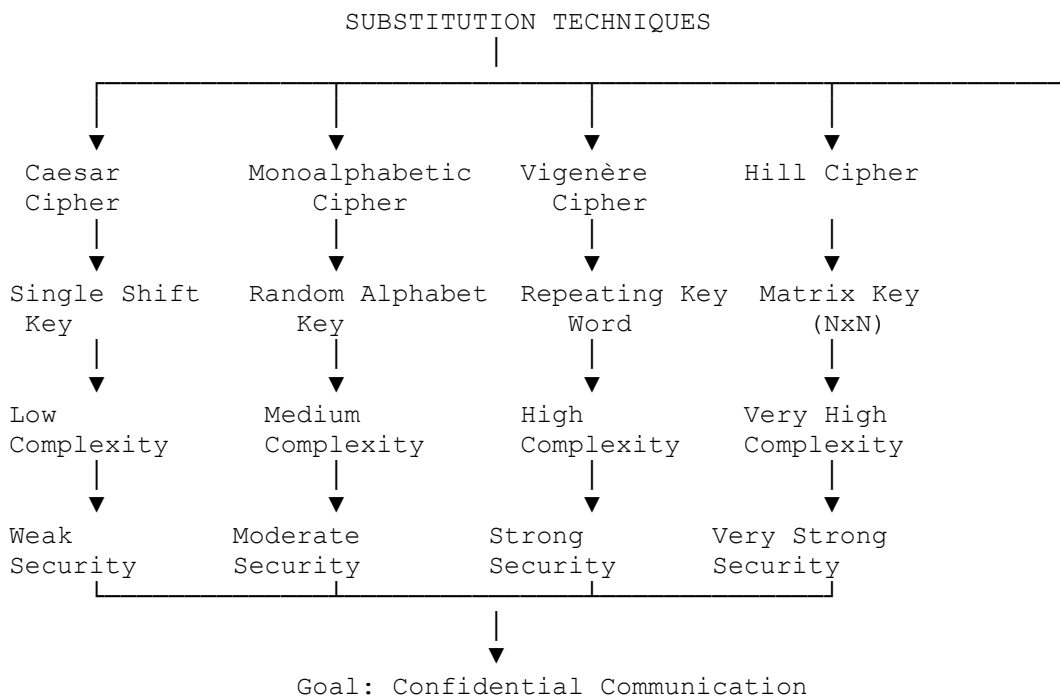
Question 2: Concept Map – Comparison of Substitution Techniques

Substitution ciphers are classical encryption techniques in which each letter of the plaintext is replaced by another letter or group of letters according to a specific rule or key. These techniques improve confidentiality by hiding the original message from unauthorized users. Different substitution ciphers vary in terms of **key usage, complexity, and security strength**.

The four commonly used substitution techniques are:

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

Concept Map (Diagram)



Comparison of Substitution Techniques

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single fixed shift key	Low	Weak
Monoalphabetic Cipher	One random substitution alphabet	Medium	Moderate
Vigenère Cipher	Repeating keyword	High	Strong
Hill Cipher	Matrix key (2×2, 3×3, etc.)	Very High	Very Strong

Explanation

1. Caesar Cipher

Key Usage

- Uses a **single shift key** (for example, Shift = 3).

Complexity

- Very simple algorithm.

Security Strength

- Weak because only **25 possible keys** exist.
- Easily broken using brute-force attacks.

Example

Plaintext: HELLO

Shift = 3

Ciphertext: **KHOOR**

2. Monoalphabetic Cipher

Key Usage

- Uses one randomly arranged alphabet.

Example

Plain Alphabet

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Cipher Alphabet

QWERTYUIOPASDFGHJKLZXCVBNM

Complexity

- More complex than Caesar Cipher.

Security Strength

- Stronger than Caesar.
- Still vulnerable to frequency analysis.

3. Vigenère Cipher

Key Usage

- Uses a **repeating keyword**.

Example

Plaintext

HELLO

Keyword

KEYKE

Complexity

- High because every letter uses a different shift.

Security Strength

- Stronger than Monoalphabetic Cipher.
- Difficult to break without knowing the keyword.

4. Hill Cipher

Key Usage

- Uses a matrix as the encryption key.

Example

Key Matrix

```
| 3 3 |  
| 2 5 |
```

Complexity

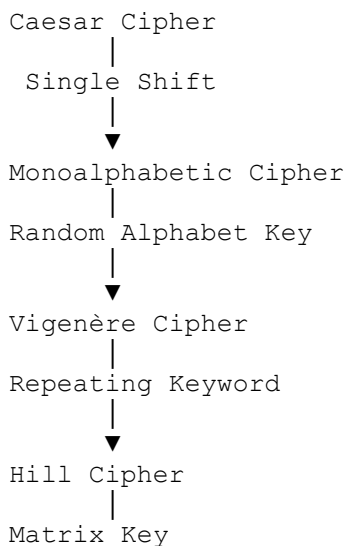
- Very high because matrix multiplication and modular arithmetic are used.

Security Strength

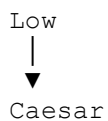
- Strongest among the four classical substitution ciphers.
- Encrypts blocks of letters instead of one letter at a time.

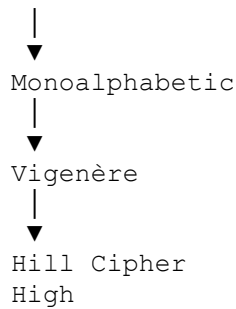
Relationship Based on Key Usage

Key Usage

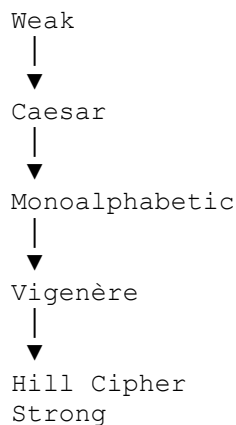


Relationship Based on Complexity





Relationship Based on Security Strength



Analysis: Which Technique is Most Secure?

Among these four techniques, the **Hill Cipher** is the **most secure** because:

- It uses **matrix keys**, making encryption more complex.
- It encrypts **blocks of letters** rather than individual letters.
- Frequency analysis becomes much more difficult.
- It combines linear algebra and modular arithmetic, increasing resistance to classical attacks.

However, compared with modern encryption algorithms such as **AES**, even the Hill Cipher is not considered secure enough for today's communication systems.

Advantages of Each Technique

Caesar Cipher

- Easy to learn and implement.
- Suitable for beginners.

Monoalphabetic Cipher

- More secure than Caesar.
- Uses a larger key space.

Vigenère Cipher

- Uses multiple alphabets.
- Stronger against frequency analysis.

Hill Cipher

- Encrypts multiple letters at once.
- Highest security among classical substitution ciphers.

Conclusion

The comparison of **Caesar Cipher**, **Monoalphabetic Cipher**, **Vigenère Cipher**, and **Hill Cipher** shows that security increases as the complexity of the key and encryption process increases. The **Caesar Cipher** is the weakest because it uses only a single shift key, while the **Hill Cipher** is the strongest because it uses matrix-based encryption and block processing. Therefore, among the given classical substitution techniques, the **Hill Cipher provides the highest level of security** due to its mathematical complexity and resistance to classical cryptanalysis.

Question 3: Concept Map – Classical Encryption Schemes (Substitution Techniques and Transposition Techniques)

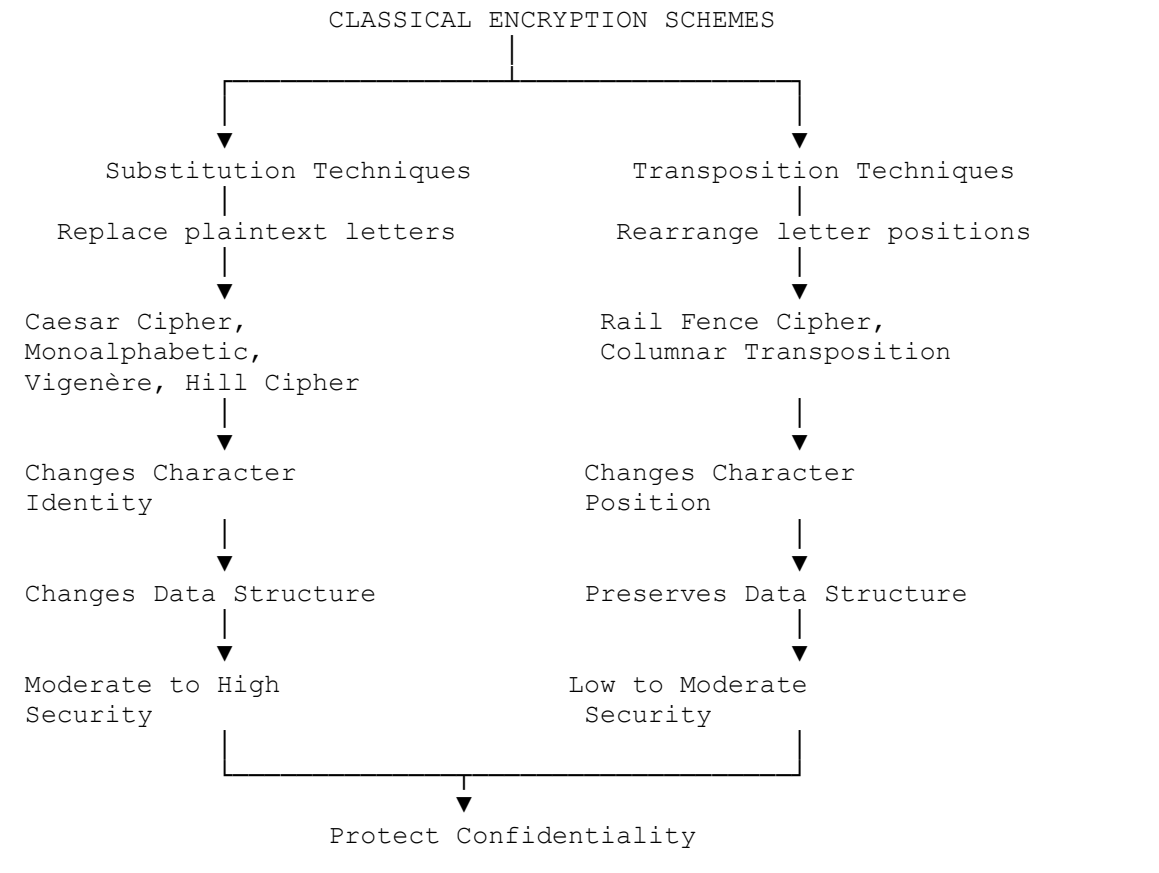
Introduction

Classical encryption schemes are traditional methods of securing information before the development of modern cryptography. They are mainly divided into two categories:

1. **Substitution Techniques** – Replace each plaintext letter with another letter or symbol.
2. **Transposition Techniques** – Rearrange the positions of the plaintext letters without changing the letters themselves.

Both techniques aim to provide confidentiality, but they differ in the way they transform the plaintext and in their level of security.

Concept Map (Diagram)



Explanation of the Concept Map

1. Classical Encryption Schemes

Classical encryption methods protect information by transforming plaintext into ciphertext. They are classified into substitution and transposition techniques.

2. Substitution Techniques

Definition

Substitution techniques replace each character in the plaintext with another character according to a predefined key.

Examples

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

Data Structure

- Changes the identity of characters.
- Original letters are replaced with different letters.

Example

Plaintext:

HELLO

Ciphertext (Caesar Cipher):

KHOOR

The letters change completely.

3. Transposition Techniques

Definition

Transposition techniques do not replace characters.

Instead, they rearrange the positions of characters.

Examples

- Rail Fence Cipher
- Columnar Transposition Cipher

Data Structure

The same letters remain present.

Only their positions change.

Example

Plaintext

HELLO

Possible Transposition

LEHLO

The letters remain the same but appear in a different order.

Similarities

Substitution Techniques

Transposition Techniques

Encrypt plaintext

Encrypt plaintext

Require an encryption key

Require an encryption key

Produce ciphertext

Produce ciphertext

Aim to provide confidentiality

Aim to provide confidentiality

Used in classical cryptography

Used in classical cryptography

Differences

Feature	Substitution Techniques	Transposition Techniques
Working Method	Replace characters	Rearrange characters
Character Identity	Changed	Not changed
Character Position	Usually remains	Changes
Data Structure	Modified	Preserved
Examples	Caesar, Vigenère, Hill	Rail Fence, Columnar

Feature	Substitution Techniques	Transposition Techniques
Security	Higher	Lower

(i) Effect on Data Structure and Security Level

Substitution Techniques

Effect on Data Structure

- Character identities are changed.
- Makes the original message more difficult to recognize.

Security Level

- Medium to High.
- Depends on the algorithm.
- Hill Cipher provides the highest security among classical substitution techniques.

Transposition Techniques

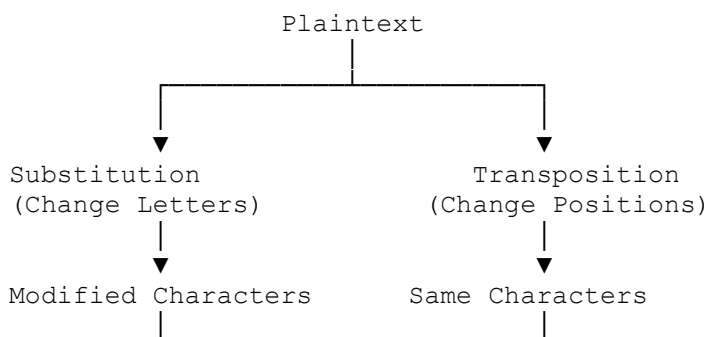
Effect on Data Structure

- Character positions change.
- Character identities remain unchanged.

Security Level

- Low to Moderate.
- Easier to analyze because letter frequencies remain unchanged.

Relationship Diagram



▼
Higher Security

▼
Lower Security

(ii) Analysis: Which Technique is More Vulnerable and Why?

Transposition techniques are generally more vulnerable to attacks than substitution techniques.

Reasons

1. The original letters remain unchanged.
2. Letter frequency is preserved.
3. Statistical analysis can easily identify patterns.
4. Brute-force attacks can recover the original arrangement quickly.

Why Substitution is Better

- Character identities are changed.
- Frequency analysis becomes more difficult.
- Advanced substitution methods such as the Hill Cipher provide better protection.

Real-World Examples

Substitution Technique

Military communication in historical times used substitution ciphers to hide sensitive messages.

Transposition Technique

Early diplomatic messages used Rail Fence and Columnar Transposition ciphers for secret communication.

Conclusion

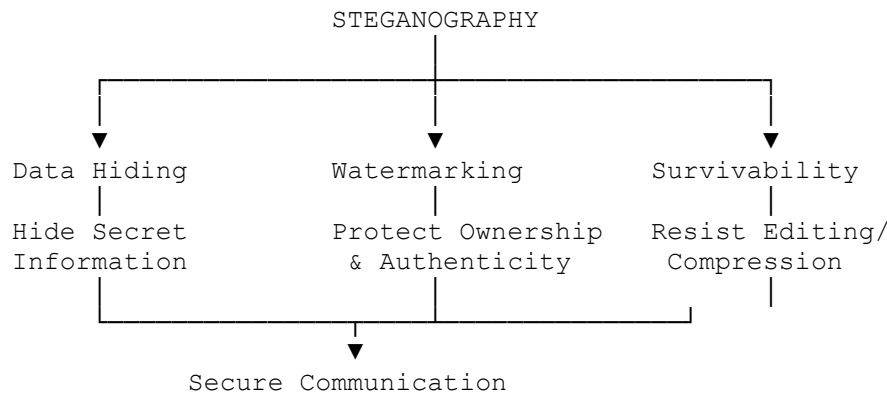
Classical encryption schemes are divided into **Substitution Techniques** and **Transposition Techniques**. Substitution techniques improve security by replacing characters with different symbols, while transposition techniques only rearrange character positions. Although both provide confidentiality, **transposition techniques are more vulnerable** because they preserve letter frequencies and are easier to break using statistical analysis. Therefore, **substitution techniques—especially the Hill Cipher—offer a higher level of security among classical encryption methods.**

Question 4: Concept Map – Steganography Concepts

Introduction

Steganography is the technique of hiding secret information inside digital media such as images, audio, or videos. It helps protect confidential information by concealing the existence of the message.

Concept Map (Diagram)



(a) Relationship

- **Data Hiding** hides confidential information inside a digital file.
- **Watermarking** protects ownership and authenticity of digital content.
- **Survivability** ensures that hidden data remains intact even after compression or editing.

(b) Contribution to Secure Communication

- **Data Hiding** keeps the message invisible to attackers.
- **Watermarking** prevents unauthorized copying and proves ownership.
- **Survivability** allows hidden information to remain available after file modifications.

(c) Real-World Applications

Concept	Real-World Application
Data Hiding	Secret military communication
Watermarking	Copyright protection for digital images and videos
Survivability	Secure sharing of images on social media

Conclusion

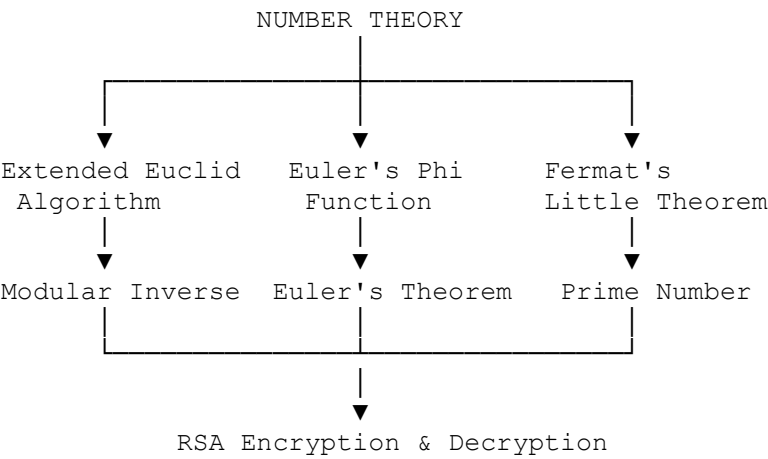
Steganography improves secure communication by hiding confidential data, protecting digital ownership, and ensuring that hidden information survives editing or compression.

Question 5: Concept Map – Number Theory in Cryptography

Introduction

Number theory provides the mathematical foundation for modern cryptography, especially public-key algorithms such as RSA.

Concept Map (Diagram)



(a) Mathematical Connection

- **Extended Euclid Algorithm** computes the **Modular Inverse**.
- **Euler's Phi Function** is used in **Euler's Theorem**.
- **Fermat's Little Theorem** is a special case of Euler's Theorem for prime numbers.

(b) Role in Encryption/Decryption

Concept	Role
Modular Inverse	Calculates private keys
Extended Euclid Algorithm	Finds modular inverse
Fermat's Little Theorem	Simplifies modular arithmetic

Concept	Role
Euler's Phi Function	Generates RSA keys
Euler's Theorem	Supports RSA encryption and decryption

(c) Most Critical Concept

Euler's Phi Function is the most important because it is used to generate RSA public and private keys. Without the Phi function, RSA key generation is not possible.

Conclusion

Number theory concepts work together to make public-key cryptography secure. Among them, **Euler's Phi Function** plays the most important role because it is essential for RSA key generation and secure encryption/decryption.